PALEOMAGNETIC DATA AND THE ACCRETION OF THE GUERRERO TERRANE, SOUTHERN MEXICO CONTINENTAL MARGIN

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Abstract. The southern Mexico continental margin has characteristics of a truncated composite margin for which both right- and left-lateral displacements of major blocks such as Baja California peninsula and northern Central America have been suggested. We present detailed paleomagnetic results for more than 1000 samples from over 100 sites in different intrusive, volcanic, and sedimentary units exposed along the southern Mexico continental margin. The data are used to study the accretion history of the Guerrero terrane and subsequent movements of the margin. Results show evidence of a complex evolution in which some areas have experienced localized tectonic rotations of varying degree but no significant latitudinal displacements. The Guerrero terrane very likely originated as a series of island arcs which were later accreted to the margin without significant latitudinal change. The accretion process resulted in deformation of the volcano-sedimentary assemblage, local rotations and variable low-degree metamorphism.

Introduction

Recent studies of the western continental margin of North America have documented a complex evolution, in which the margin is composed of several blocks or terranes with distinct tectonic and stratigraphic records, implying a highly mobilistic history (e.g. Beck, 1976, 1980, 1988; Irving, 1979). Prior to accretion to the continental margin, such allochthonous terranes may have been transported by two main processes: by riding on an oceanic plate or by oblique convergence of detached fragments along a subduction trench or a transform fault system (Beck, 1983). Both processes may cause changes in geographical latitude and/or rotation of terranes, resulting in remanence inclinations and declinations which are in discordance with those of coeval neighbouring autochthonous rocks.

Campa and Coney (1983) analyzed the (still sparse) geological and tectonic information for Mexico to construct a preliminary map of tectonostratigraphic terranes. Terrane boundaries and detailed subdivisions were difficult to define, as large parts of Mexico are overlain either by Tertiary and Quaternary volcanic rocks or Cretaceous limestones. In their map most of the Pacific margin of Mexico is represented by the Guerrero and Xolapa terranes (Fig. 1). The Guerrero terrane is characterized by magmatic, volcaniclastic, and sedimentary rocks typical of island arcs. Campa and Coney (1983) assumed that this terrane was accreted to Mexico during the Laramide orogeny in Late Cretaceous to Early Tertiary time. The Xolapa terrane consists of mainly metamorphic rocks of rather uncertain age, probably representing the roots of a Mesozoic magmatic arc. Accretion possibly took place between Middle Jurassic and Early Tertiary (Moran-Zenteno et al., 1986).

Southern Mexico Continental Margin

The margin between Puerto Vallarta and Acapulco (Fig. 1) has been divided into the Guerrero and Xolapa terranes (Campa and Coney, 1983). The area studied further includes the Mixteco terrane, which differs from the two other terranes in the much older age and metamorphic nature of its basement.

Guerrero Terrane

The variety of rock types, stratigraphy, and structural styles of the large Guerrero terrane indicate its composite nature. Campa and Coney (1983) identified three separate subterranes, Teloloapan-Ixtapan, Zihuatanejo and Huetamo. How-

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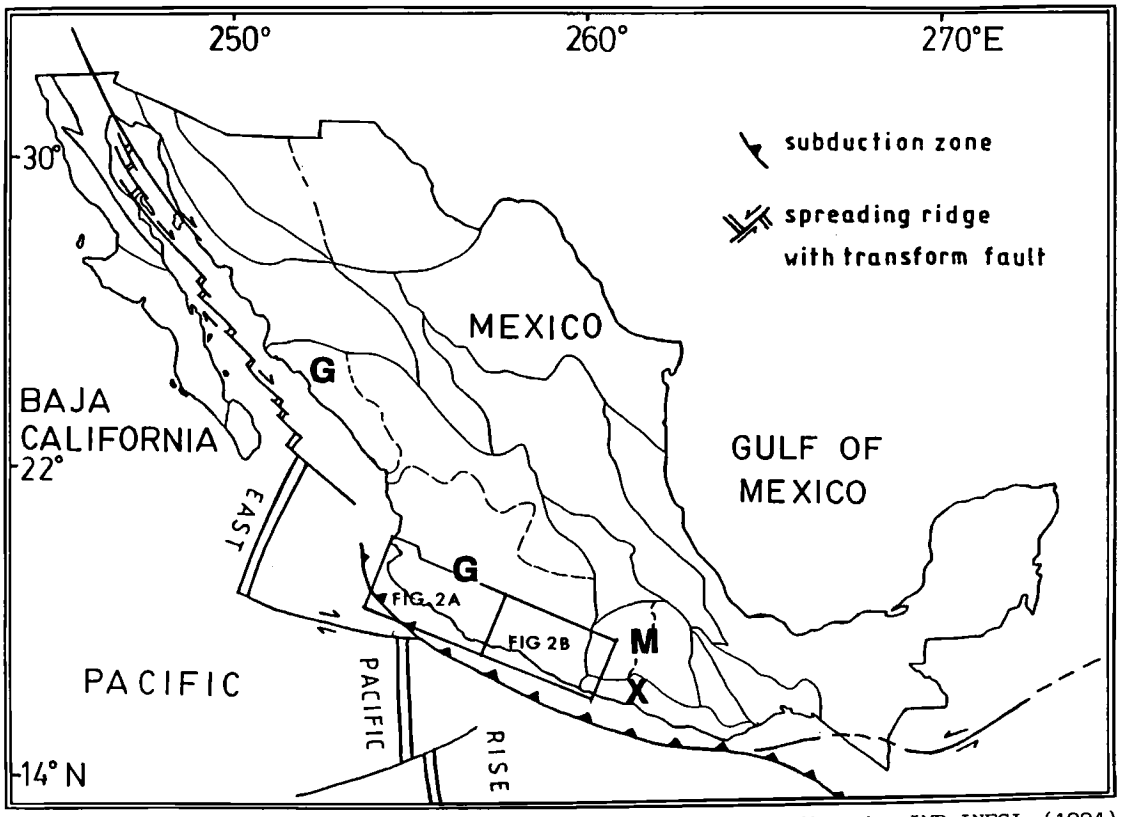


Fig. 1. Tectonostratigraphic map for Mexico according to IMP-INEGI (1984), based largely on map given by Campa and Coney (1983). G, X, and M indicate the Guerrero, Xolapa, and Mixteco terrane, respectively. Dashed lines indicate terrane subdivisions.

ever, the extension and characteristics of these subterranes are not well defined, nor are the boundaries with the adjacent subterranes and terranes. The terrane is in general characterized by submarine volcanic and sedimentary sequences of Jurassic to mid-Cretaceous ages. A Pb-α radiometric date of 315 ±15 Ma from zircons in an intrusive body reported by Grajales-Nishimura and Lopez-Infanzon (1983), which suggested there are remnants of older basement rocks in the terrane, was not supported by the Rb/Sr age of 69 ±1 Ma obtained by us.

The Zihuatanejo subterrane extends along the coast south of Puerto Vallarta and includes mainly andesitic volcanics and interbedded Albian limestones. There are also some shales, sandstones, conglomerates, and some red beds with dinosaur footprints (Ferrusquia-Villafranca et al., 1978). The assemblage is deformed but not significantly metamorphosed (Campa and Coney, 1983). Three PEMEX ('Petroleos Mexicanos') wells (see Grajales-Nishimura and Lopez-Infanzon, 1983) in the area south of Colima City indicate differences in the stratigraphy which may reflect changing environments of relatively small basins and island arcs. Pantoja-Alor (1983, 1988, personal communication) reviewed the available radiometric dates and indicated a plutonic event beginning at the end of the Albian and lasting until the Coniacian. The magmatic activity resumed during the Tertiary. Most of the paleomagnetic samples come from intrusive bodies, volcanics, and limestones in this subterrane.

The Huetamo subterrane is characterized by Upper Jurassic marine volcaniclastic sediments overlain by a Neocomian sequence of sands and shales. This flysch-like sequence is covered by Albian limestones interbedded with tuffs and red beds. They are covered by abundant Upper Cretaceous (?) red beds and ignimbrites. The assemblage is deformed but not metamorphosed.

The Teloloapan-Ixtapan subterrane is characterized by a sequence of andesitic volcanics and volcaniclastic sediments with interbedded limestones, shales and sandstones. The assemblage shows variable low-grade regional metamorphism and has been severely deformed.

Along the margin there are several large calc-alkaline batholiths emplaced within the complex

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sedimentary and metamorphic sequence of the Guerrero terrane. Igneous activity may be as old as Jurassic (Pantoja-Alor, 1983; De Cserna, 1978; Mujica, 1978; Pantoja-Alor and Estrada-Barraza, 1986; Köhler et al., 1988). During the Late Jurassic and up to the Early Cretaceous, calc-alkaline magmatic activity, mainly submarine and of andesite-dacite composition, occurred along the margin. Polarity of subduction has been discussed, and both an eastward subduction zone (Campa, 1978) and a westward subduction zone (Urrutia-Fucugauchi, 1980) have been proposed.

Isotopic, geochemical, and petrographic studies of intrusive bodies along the margin are strongly suggestive of a magmatic arc environment (e.g. Köhler et al., 1988). Granites have a calc-alkaline to highly calc-alkaline character and contain biotite, and hornblende, plus magnetite, and titanomagnetites. The granites correspond to the I-type of Chappel and White (1974) and to the magnetite series of Ishihara (1977) and Takahashi et al. (1974). Detailed opaque mineral studies of the granitoid complexes at Acapulco, Xaltianguis and Ocotito (Böhnel et al., 1988) and Puerto Vallarta (Böhnel and Negendank, 1988) show that they contain oxides of the titanomagnetite series with varying grain sizes and high temperature oxidation states. Initial strontium ratios for the Puerto Vallarta Complex are between 0.704 to 0.706, supporting a possible mantle origin (Köhler et al., 1988). The intrusion age of the complex is set around 105 Ma as calculated from 20 Rb/Sr whole-rock determinations. Rb/Sr ratios for hornblende and biotite separates yielded cooling ages of 86 ±2 Ma and 83 ±3 Ma, respectively. The elongated mineral belts roughly parallel to the margin (Fe-Cu and Au-Mo-Ag-Pb-Zn-Sn-Hg) correspond well with the distribution of mineral deposits related to plate subduction processes (e.g. Sillitoe, 1972).

Volcanic rocks of Cretaceous to Tertiary age are found in the Tecalitlan, El Encino, Mal Paso and Xochipala Formations, and in many other areas of the States of Jalisco, Colima, Michoacan, and Guerrero.

At the end of the Albian, several calc-alkaline plutons were emplaced in the Guerrero terrane. The composition varies from granodiorite to quartz-monzonite with high contents of K and low strontium ratios. This is followed by an apparent hiatus from the end of the Coniacian to the Maastrichtian or early Tertiary, correlating with the Laramide orogeny. Magmatic activity resumed during the Paleocene and continued up to the Late Oligocene (Pantoja-Alor, 1983). Volcanic activity of a more acid character is represented in the Balsas, Chacharando, Papagayo, Agua de Obispo, Alquitran, Tilzapotla, and Buenavista Formations. This volcanic activity may be similar to the ignimbrites of the Sierra Madre Occidental provinc in northwestern Mexico, which may suggest a continuity of the magmatic arc and subduction zone along the entire margin. Post-Miocene volcanic activity is apparently restricted to areas north of the margin in the Trans-Mexican Volcanic Belt, which includes the historic and present-day volcanic activity of central-southern Mexico related to subduction of the Cocos plate along the Middle America trench.

Mixteco Terrane

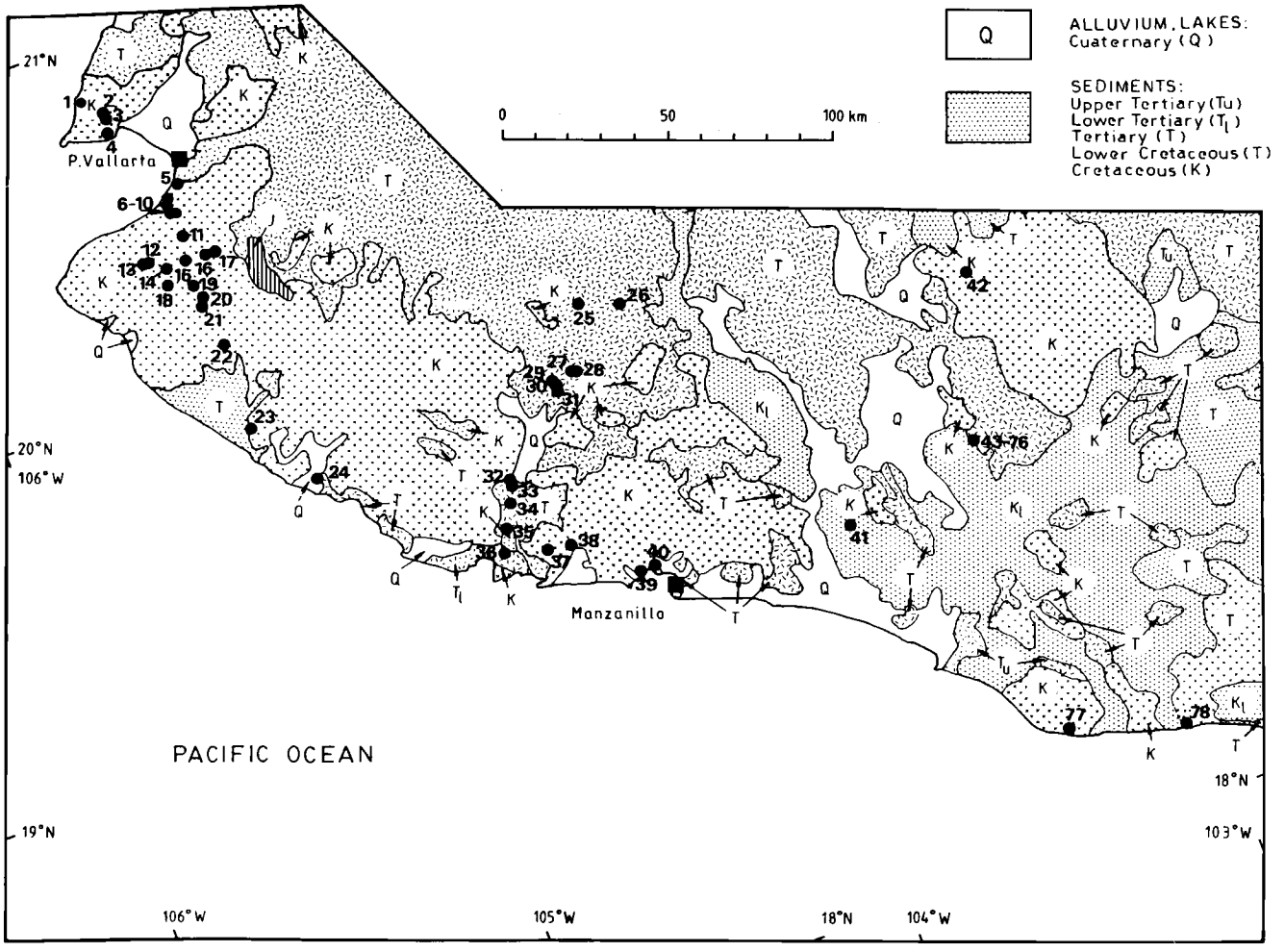
The Mixteco terrane is characterized by a Paleozoic basement grouped into the Acatlan complex (Ortega-Gutierrez, 1981). This complex is composed of a para-autochthonous association of meta-sediments, meta-volcanics, and ultramafic intrusives metamorphosed to shists and migmatites, and by an allochthonous association of meta- ophiolites, greenstones, mylonites, and a quartz-clastic unit. Radiometric ages range from 480 Ma to 173 Ma (Ruiz-Castellanos, 1979), but this age range may extend to 540 Ma as indicated by the presence of crinoids. The Acatlan Complex is intruded by the Ordovician trondhjemitic Totol- tepec Stock, and a swarm of Triassic undeformed granitic, aplitic, and pegmatitic sills and dikes, the San Miguel dikes (Fries et al., 1970; Ruiz-Castellanos, 1979; Ortega-Gutierrez, 1981). The Acatlan Complex presents a complex structure as a result of at least four phases of deformation, most of them noncoaxially (Ortega-Gutierrez, 1981).

Paleomagnetic studies are being completed in areas to the east of the region discussed in this paper (e.g. Fang et al., 1986; Moran-Zenteno et al., 1986). The paleomagnetic data seem to reflect the open folding structural pattern characteristic of the Acatlan, and the effects of remagnetization related to the various phases of deformation and igneous intrusions.

Xolapa Terrane

The Xolapa Complex extends along a belt about 600 km long and some 50-150 km wide on the continental margin, to the southeast of Zihuatanejo. In the area north of Acapulco, the Xolapa terrane is composed of biotite schists and gneiss, quartzites, marble and pegmatites (De Cserna, 1965). To the southeast, there are greenschists (from sedimentary rocks) and orthogneiss and migmatites. The age and stratigraphy of the Xolapa Complex are uncertain. Radiometric dates range from the Cambrian to the Tertiary (e.g. De Cserna et al., 1962, 1975; Halpern et al., 1974; Guerrero-Garcia et al., 1978). Precambrian and Paleozoic ages have also been proposed based on geological inferences (e.g. De Cserna, 1965; Kesler and Heath, 1970). Granitic intrusions of varying compositions from granite, quartz-monzonite and granodiorite of Late Cretaceous to Early Tertiary ages are also present. Ortega-Gutierrez (1981) has interpreted the complex as the roots of a Mesozoic magmatic arc. Moran-Zenteno et al. (1986) have interpreted that the Xolapa terrane may have arrived at its present

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(a)

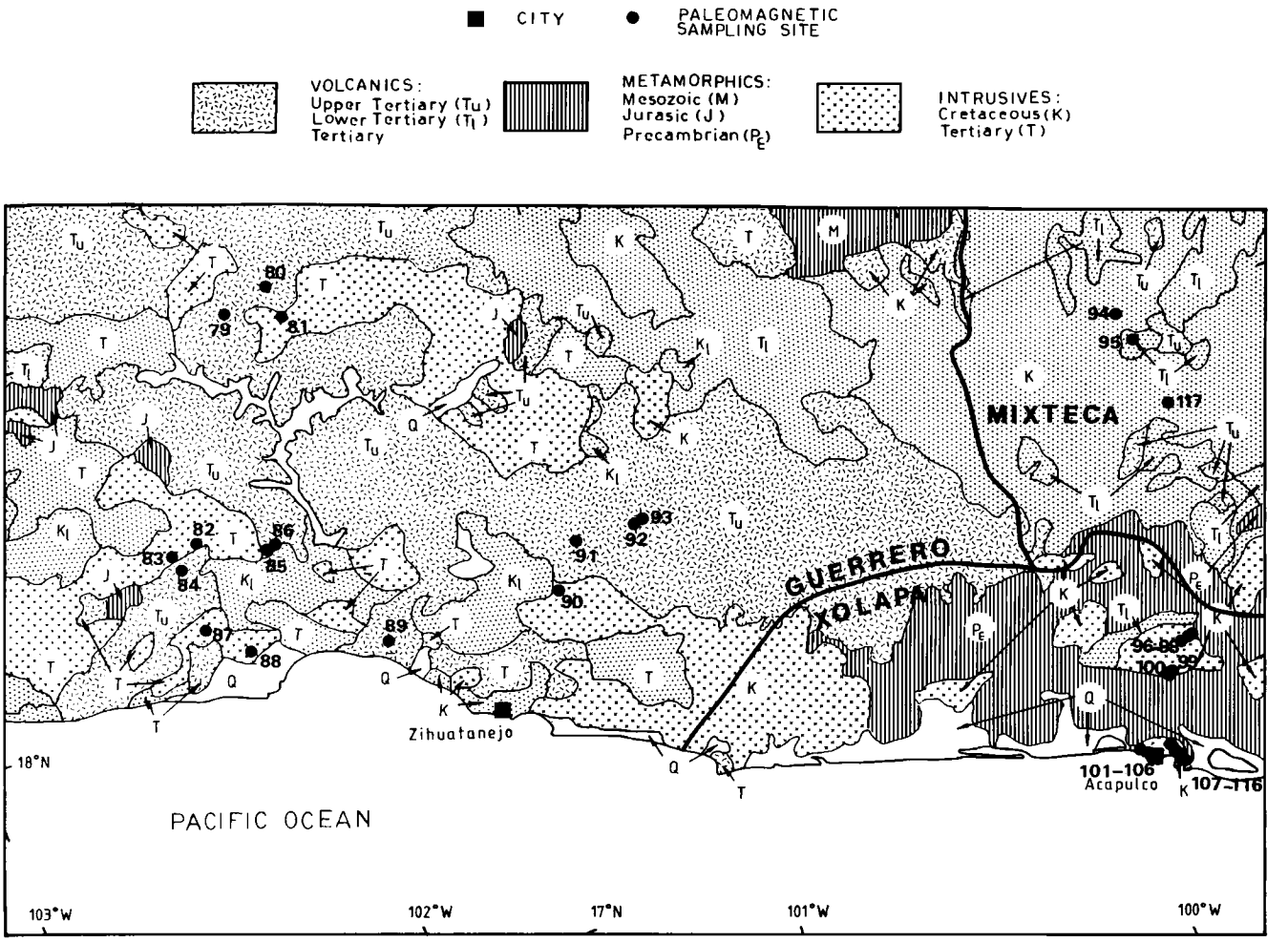
Fig. 2. Simplified geological map of the study area : (a) northwestern part, and (b) southeastern part (see Fig. 1). Paleomagnetic sampling sites are indicated with circles and code numbers referring to Tables 1 and 2. The boundaries between the Guerrero, Xolapa, and Mixteco terranes are indicated by thick lines.

relative position sometime from the Middle Jurassic and the Late Cretaceous-Early Tertiary.

Paleomagnetic Sampling

For the paleomagnetic studies, about 1000 oriented samples from several intrusive, volcanic and sedimentary units exposed along the southern Mexico continental margin (Fig. 2) were collected from more than 100 sites (Table 1). Sampling was carried out with a portable drill with non-magnetic diamond-impregnated drill bits and orientations were determined with magnetic compass. Most sites are road cuts or exposures along rivers as sampling was difficult because of profound weathering and limited access. Four regions were studied in more detail: the Puerto Vallarta intrusive complex with 22 sites (No. 1 to 22). These granitoids rocks were emplaced around 105 Ma (Rb/Sr whole rock age) ago, and give cooling ages as old as 80 Ma (Rb/Sr and K/Ar dates, Gastil et al., 1976; Grajales-Nishimura and Lopez-Infanzon, 1983; Köhler et al., 1988; own data). Near Manzanillo, Lower Tertiary to Upper Cretaceous volcanic and intrusive rocks were sampled at 16 sites (No. 25 to 40). According to whole rock Rb/Sr and K/Ar determinations the igneous rocks were emplaced between about 40 Ma and 70 Ma (Pantoja-Alor, 1983, 1988; own data). The Encino mining district was subject of an aeromagnetic - paleomagnetic study (Alva-Valdivia et al., in press). Samples were taken from 35 sites (No. 43 to 76) which belong to either the Aptian - Albian or the Conacian - Maastrichtian. Finally, volcanic and intrusive rocks

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(b)

Fig. 2. **Continued**

of predominantly Upper Cretaceous to Lower Tertiary age were studied at 12 sites near Zihuatanejo (No. 82 to 93). From the Xolapa and Mixteco terranes results are already published for the Acapulco granite (No. 101 to 116), Xaltianguis monzonite (No. 98 to 100), and the Morelos Formation in two regions (see Table 2) (Urrutia-Fucugauchi, 1983; Böhnel, 1985; Böhnel et al., 1988; Urrutia-Fucugauchi, 1988).

Rock Magnetic Studies and Ore Petrology

The direction and intensity of the natural remanent magnetization (NRM) were measured with Digico spinner magnetometers. The magnetic stability and vectorial composition of NRM were investigated by detailed step-wise thermal or alternating field (AF) demagnetization. Thermal demagnetization was carried out in 8-10 steps from room temperature up to 670°C in two non-inductive electric furnaces (a TSD-1 Schonstedt and an apparatus constructed at the University of Münster, FRG). AF demagnetization was made in 8-11 steps up to 300 mT in two AF demagnetizers (a three-axis Schonstedt model and an apparatus made at the University of Münster, FRG).

Rock-magnetic measurements for identification of magnetic carriers, including acquisition of isothermal remanent magnetization (IRM), Curie temperature determination, IRM variation with temperatures down to 77 K, and low-field susceptibility, were also completed on selected samples. These studies revealed important details about the origin and possible alterations of the remanent magnetization. Details are given by Böhnel et al. (1988), Böhnel and Negendank (1988) and Alva-Valdivia et al. (1988) and will be summarized here only briefly.

Site-mean values of remanence intensity, low-field susceptibility and Königsberger ratio

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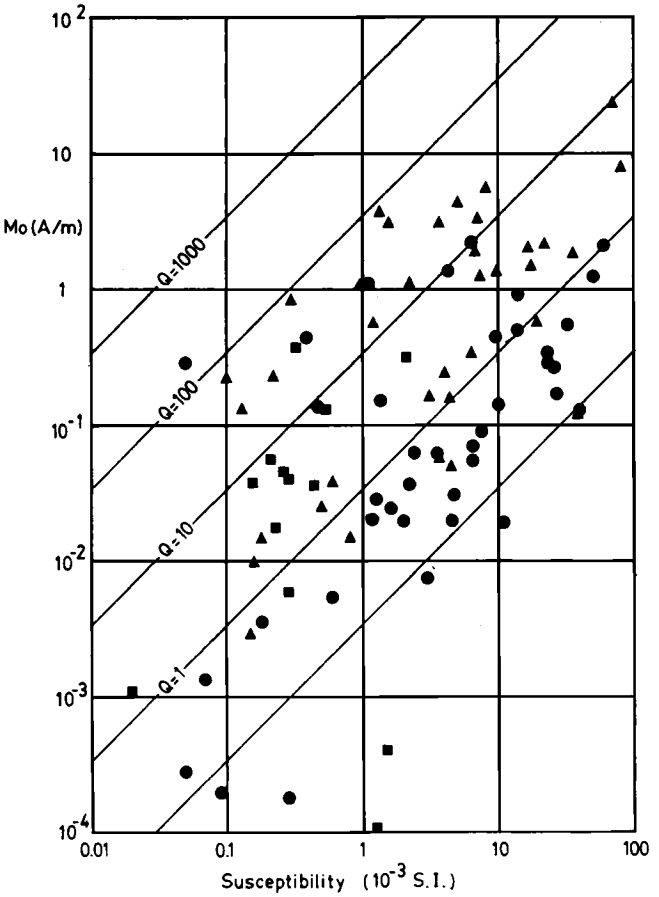


Fig. 3. Site mean values for NRM-intensity and susceptibility (SI unit system); lines of constant Königsberger ratio Q. Circles: intrusive rocks; triangles: volcanic rocks; squares: sediments. Massive iron ores are not included in the diagram.

(Q) for the different lithologies are given in Table 2. Susceptibility ranges from about $0.16 \cdot 10^{-3}$ for limestones and altered volcanics and intrusives to about $580 \cdot 10^{-3}$ (SI unit system) for magnetite-bearing iron ores. Intensity and susceptibility vary within a given site and among sites in a given lithology (Fig. 3). Nevertheless, there is a general correlation of high NRM intensity with high susceptibility, in particular for the intrusive rocks, which may be explained by variations in the volume content of (titano-) magnetite. The Königsberger tratio (Q) also shows large variations, which are likely due to differences in grain size and chemical composition of remanence carriers, rather than to varying volumes of magnetic minerals. For intrusive rocks Q has values mainly between 0.1 and 1, whereas Q covers a much larger range between about 1 and 100 for volcanic rocks. Obviously in the volcanics, variations of magnetic mineralogy and grain size spectra are much more variable than in the intrusive rocks.

From several sites thermomagnetic curves were obtained by measuring the induced magnetization in a strong direct magnetic field as a function of temperature. Intrusive rocks present almost reversible curves with Curie temperatures near 580°C, indicating magnetite as the dominant magnetic mineral. The volcanic rocks generally exhibited irreversible curves with Curie temperatures reaching up to 630°C. This may be due to high- and low-temperature and hydrothermal alteration of the minerals, resulting in the formation of titanohematites and -maghemites.

Low temperature variations of saturation isothermal remanent magnetization (SIRM) down to 77 K support these interpretations. The SIRM temperature dependence for volcanic rocks is similar to that reported for titanomagnetites of varying high- and low-temperature oxidation state (Dankers, 1978; Harstra, 1982). Intrusive rocks show a distinct decrease of SIRM with temperature, especially around 120 K, but only small to moderate recovery of remanence during reheating. Here bulk magnetic properties are dominated by magnetite or Ti-poor titanomagnetite of large grain size (Levi and Merrill, 1978).

IRM acquisition experiments were carried out with volcanic rocks and limestones in fields up to 1.5 T. Volcanic rocks often exhibited two components with different coercivity spectra. One component saturates below 0.1 T, which is typical for magnetite (Dunlop, 1972). The other component saturates in fields between 0.5 and 1 T, probably due to the presence of small titanomagnetite or titanohematite particles. These may have originated by exsolution during high-temperature oxidation and by alteration of silicate minerals. Bulk magnetic properties of limestone samples seem to be dominated by (titano-) magnetite, as they reach saturation in fields below 0.1. Only few samples saturated in fields up to 0.5 T, which may be due to a goethite or hematite component.

The coercive force H_c is low for intrusive rocks with values around 4 mT. The ratio of remanent saturation magnetization to saturation magnetization $M_{rs}/M_s \leq 0.1$, indicating an important contribution of multidomain particles (Day et al., 1977).

Polished sections were prepared from representative samples and observed in reflected light. We also used a scanning electron microscope, which allowed determination of the atomic composition (for atomic weight numbers greater than 20). Volcanic rocks are characterized by highly altered opaque minerals as result of high- and low-temperature oxidation and hydrothermal processes. Intrusive rocks, in contrast, contain rather fresh Ti-poor titanomagnetites, showing only minor effects of maghemitization. The magnetic grains in these rocks are much greater than the grains in volcanic rocks and ranged between 20 μm and 1 mm. In most intrusive rocks pyrite was found, often containing traces of titanium. Probably this points to very small titanomagnetite inclusions, which could easily retain a primary magnetization.

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TABLE 1. Paleomagnetic Sampling Sites in the Guerrero, Xolapa, and Mixteco Terranes, Southern Mexico.

Site	Coordinates deg N	deg W	Rock Type	Age and Comments
1	20.86	105.42	Granitoid rocks	According to several geochronological studies
2	20.81	105.39	from the Puerto	(see references a, c, d, and f), the rocks of
3	20.80	105.38	Vallarta Batholith	this intrusive complex were emplaced around
4	20.75	105.40	(granites, grano-	105 Ma ago (20 Rb/Sr whole rock determinations).
5	20.57	105.25	diorites, diorites,	Mineral cooling ages (Rb/Sr and K/Ar biotite
6	20.54	105.29	gabbros); exposed	determinations) extend to 80 Ma. A cooling rate
7	20.53	105.30	in roadcuts and	of ca. 35 °C/Ma was proposed by Köhler et al.
8	20.51	105.31	along rivers. Most	(1988). Therefore, reemanence acquisition most
9	19.46	104.66	of the samples are	probably occurred between 100 Ma and 80 Ma ago.
10	20.48	105.29	fresh; north of	
11	20.41	105.30	Puerto Vallarta	
12	20.38	105.42	weathering was pro-	
13	20.38	105.42	found and access	
14	20.35	105.37	limited, so that	
15	20.35	105.32	here the rocks are	
16	20.35	105.26	more altered.	
17	20.35	105.23		
18	20.31	105.39		
19	20.28	105.32		
20	20.24	105.31		
21	20.24	105.32		
22	20.09	105.30		
23	19.84	105.31	Volcanite	Early Tertiary
24	19.66	105.18	Volcanite	Early Tertiary
25	19.87	104.29	Basaltic andesite	Tertiary
26	19.82	104.18	Andesites	Cretaceous
27	19.70	104.37	Volcanites	Tertiary
28	19.70	104.38	(dacites to	
29	19.69	104.42	basaltic	
30	19.68	104.42	andesites)	
31	19.66	104.44		
32	20.49	105.30	Granodiorite	Late Cretaceous
33	19.46	104.66	Gabbro	Early Tertiary
34	19.41	104.67	Rhyolite	Tertiary
35	19.35	104.72	Limestone	Early Cretaceous
36	19.29	104.73	Andesite	Early Tertiary
37	19.26	104.73	Andesite	Tertiary
38	19.25	104.55	Granite w. xenolites	69 ±1 Ma (f)
39	19.11	104.38	Diorite	Late Cretaceous
40	19.12	104.33	Gabbro	Late Cretaceous
41	19.04	103.79	Limestone	Early Cretaceous
42	19.56	103.20	Gabbro	53 ±1 Ma (e) for the intrusive complex
43	19.14	103.36	Vulcanite & Fe-ore	Maastrichtian Sites 43 to 76 are from El Encino
44	19.14	103.36	Diorite & Fe-ore	Maastrichtian Encino area, which is an iron-
45	19.14	103.36	Volcanite & Fe-ore	Maastrichtian mining district. El Encino Forma-
46	19.14	103.35	Skarn-Fe	Maastrichtian tion consists of a lower member
47	19.14	103.35	Massive Fe-ore	Maastrichtian formedby an andesitic conglome-
48	19.14	103.36	Massive Fe-ore	Maastrichtian rate with interbedded reef lime-
49	19.14	103.36	Massive Fe-ore	Maastrichtian stones and porphyritic dikes and
50	19.15	103.34	Granodiorite	Maastrichtian sills of andesitic to dacitic
51	19.15	103.34	Granodiorite	Maastrichtian composition. The upper member is
52	19.14	103.36	Andesitic breccia	Senonian formed by sandstones, limolites
53	19.14	103.36	Andesitic breccia	Senonian and interbedded limestone conglo-
54	19.14	103.36	Andesitic breccia	Senonian merate and limestones. The age of
55	19.14	103.36	Monzonite	Cenomanian the Encino is Late Aptian.
56	19.14	103.35	Andesitic tuff	Late Aptian The Vallecitos Formation is di-
57	19.14	103.34	Volcanite	Late Aptian vided into a lower tuffaceous

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TABLE 1. (continued)

Site	Coordinates deg N	deg W	Rock Type	Age and Comments
58	19.14	103.36	Cuarzomonzonite	Early Albian
59	19.14	103.34	Volcanite	Late Aptian
60	19.15	103.34	Volcanite	Late Aptian
61	19.13	103.35	Tuff	Early Albian
62	19.14	103.33	Volcanite	Early Albian
63	19.12	103.36	Volcanite	Early Albian
64	19.14	103.35	Volcanite	Early Albian
65	19.14	103.36	Ignimbrite	Early Albian
66	19.14	103.35	Sedimentary	Late Aptian
67	19.14	103.35	Sedimentary	Late Aptian
68	19.14	103.35	Sedimentary	Late Aptian
69	19.14	103.35	Limestone	Late Aptian
70	19.14	103.35	Limestone	Late Aptian
71	19.14	103.35	Sandstone	Late Aptian
72	19.13	103.36	Sandstone	Late Aptian
73	19.14	103.35	Limestone	Late Aptian
74	19.13	103.35	Conglomerate	Early Albian
75	19.14	103.36	Limestone	Early Albian
76	19.14	103.36	Limestone	Early Albian
77	18.29	103.39	Granite	K/Ar biotite age (c) : 38 ± Ma, 43 ±5 Ma
78	18.20	103.08	Granite	K/Ar biotite age (c) : 55 ±4 Ma
79	18.94	101.88	Andesite	Cretaceous
80	18.97	101.73	Basalt	Tertiary
81	18.88	101.74	Granite	Late Tertiary
82	18.41	102.21	Granitoid rocks	Intrusive complex around Arteaga; ages between
83	18.37	102.29	(monzodiorite	44 ±3 Ma and 57 ±1 Ma (c) are reported for
84	18.33	102.28	to granodiorite)	the rocks sampled for paleomagnetic studies
85	18.30	102.26		
86	18.30	102.07		
87	18.15	102.28	Tonalite	Ages of 56 ±5 and 60 ±5 were reported from
88	18.05	102.18	Andesite	this sampling region (e)
89	17.93	101.79	Volcanite	Tertiary
90	17.9	101.3	Ignimbrite	K/Ar ages between 42 ±1 Ma and 61 ±1 Ma were
91	18.0	101.2	Volcanite	reported for this sampling locality (e) (up
92	18.0	101.2	Volcanite	to site 93)
93	18.0	101.2	Granite	
94	19.02	99.54	Monzonite	
95	18.95	99.52	Andesitic breccia	Tetelcingo Formation; 67 ±3 Ma (b)
96-100	17.15	99.71	Granitoid	Xaltianguis intrusive complex; 30 ±1 Ma (f)
101-116	16.84	99.86	Granitoid	Acapulco intrusive complex; 43 ±1 Ma (f)
117	17.75	99.50	Limestone	Morelos Formation; Albian-Cenomanian

Note: The geological ages given in the last column age based on the geological maps of INEGI (1980), except for the Encino mining district, where they are based on Pantoja-Alor (1983); absolute ages are based on (a) Gastil et al. (1976), (b) Chavez-Quirarte (1982), (c) Grajales-Nishimura and Infanzon (1983), (d) Köhler et al. (1988), (e) Pantoja-Alor (1988, and personal communication, (f) our own determinations. Sites 96-116 (Bhnel et al., 1988) and 117 (Urrutia-Fucugauchi, 1988) are included for referenece.

Demagnetization Experiments

All specimens were AF or thermally demagnetized at 4 to 10 steps, according to the characterisics observed in at least 2 pilot specimens per site. Generally the directional changes of magnetization were analyzed using Zijderveld diagrams (Fig. 4) (As 1960; Zijderveld, 1967), together with equal-area plots of directions and difference vectors were also analyzed. In some cases, principal components analysis (Kirschvink, 1980) was applied to the demagnetization data.

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TABLE 2. Paleomagnetic Results from the Guerrero and Mixteco Terranes, Southern Mexico

No.	NRM A/m # $\cdot 10^{-3}$	Susc. (SI) $\cdot 10^{-3}$	Q	MDF MDT	n/r	R	k	α_{95} deg	Inc/Dec tilt uncorr. deg	Inc/Dec tilt corr. deg	Lat/Long deg
1*	20.7 #	1.20	0.4	2.8 mT	6/0	5.410	9	24.4	-18.4/320.0	-18.4/320.0	40.4/130.9
2	55.9 #	6.6	0.3	6.7 mT	6/0	5.751	20	15.3	39.9/ 8.1	39.9/ 8.1	82.2/329.2
3	0.18#	0.29	0.01								
4	5.50#	0.60	0.4	>75 mT	6/1	5.951	103	6.6	46.7/334.6	46.7/334.6	65.8/186.6
5	62.4 #	2.39	0.76	3.8 mT	4/1	3.884	26	18.6	50.7/333.7	50.7/333.7	66.1/194.8
6	37.8 #	2.44	0.53	5.3 mT	5/1	4.840	25	15.6	31.1/347.1	31.1/347.1	77.2/149.9
7	7.40#	2.96	0.07	26 mT	7/0	6.786	28	11.6	33.3/ 5.7	33.3/ 5.7	85.3/ 39.6
8	19.1 #	11.0	0.66	17 mT	5/0	4.870	31	14.0	-47.1/160.1	-47.1/160.1	-70.3/ 11.9
9	19.9 #	2.00	0.2	3.0 mT	5/0	4.979	190	5.6	32.9/ 14.9	32.9/ 14.9	80.9/344.3
10	24.1 #	1.68	0.47	12 mT	5/1	4.902	41	12.1	-36.3/170.3	-36.3/170.3	-80.9/344.3
11	88.9 #	7.65	0.34	4.8 mT	5/2	4.731	21	20.5	41.2/292.7	41.2/292.7	24.1/184.6
12	62.3 #	3.57	0.49	10 mT	7/0	6.925	80	6.8	41.7/342.6	41.7/342.6	73.5/180.4
13	0.51	14.2	1.0	81 mT	6/0	5.945	92	7.0	-26.8/180.6	-26.8/180.6	-83.7/249.2
14	0.29	0.05	201.	>500 °C	7/0	6.760	25	12.3	-43.9/164.6	-43.9/164.6	-74.9/ 8.2
15	0.56	32.6	0.52	5.6 mT	8/0	7.588	17	14.1	45.7/356.2	45.7/356.2	82.4/228.3
16	0.28#	0.05	0.15								
17	0.19#	0.09	0.06								
18	27.6 #	1.27	0.75	13 mT	8/0	7.500	14	15.1	36.3/344.3	36.3/344.3	75.3/166.8
19*	0.29	23.7	0.34	5.4 mT	4/1	3.912	34	15.9	-6.9/218.0	-6.9/218.0	-49.3/184.1
20	0.14	9.94	0.37	3.3 mT							
21*	31.2 #	4.72	0.19	4.3 mT	5/0	4.826	23	16.3	-24.1/ 37.9	-24.1/ 37.9	40.3/ 22.8
22	44.4	9.50	0.13	1.8 mT	6/2	5.286	7	27.0	41.5/342.5	41.5/342.5	73.4/180.6
23	0.23	0.22	31.2	>620 °C	6/0	5.923	65	8.4	17.7/ 45.0	17.7/ 45.0	45.2/351.9
24	24.8 #	0.54	1.36	48 mT	6/0	5.706	17	17.0	-24.2/188.0	-24.2/188.0	-79.6/206.1
25	0.24	3.93	1.73	>75 mT	7/0	6.998	3251	1.1	41.0/357.5	41.0/357.5	85.6/223.9
26	9.60#	0.16	2.5	46 mT	6/1	4.969	130	6.7	41.0/357.5	39.0/357.5	86.8/210.2
27	1.85	35.9	1.44	6.3 mT	5/1	4.964	112	7.3	45.3/306.8	29.1/331.6	62.7/161.4
28	1.50	17.6	2.76	66 mT	7/0	6.991	634	2.4	57.9/314.7	35.5/345.6	76.4/167.7
29	2.07	16.5	4.02	34 mT	7/0	6.969	191	4.4	42.9/331.1	18.4/334.3	63.1/146.9
30	0.32	6.31	1.40	314 °C	8/1	6.926	81	6.8	53.5/326.8	28.7/329.5	60.6/161.9
31	35.6 #	0.44	1.02	417 °C	4/1	3.932	44	13.9	-47.8/142.0	-24.2/148.3	-58.8/337.5
32	0.93			11 mT	7/1	5.975	203	4.7	33.3/302.7	33.3/302.7	36.0/174.1
33	1.14	1.20	21.0	86 mT	7/0	6.973	220	4.1	-49.0/132.1	-49.0/132.1	-45.6/ 8.6
34	2.90#	0.15	0.54	19 mT	5/0	4.940	67	9.4	26.5/349.1	26.5/349.1	78.3/139.6
35*	46.3 #	0.26	5.07	600 °C	9/0	8.556	18	12.7	-3.7/288.5	41.9/280.4	16.9/185.6
36*	0.16	4.42	38.1	500 °C	11/0	10.756	41	7.2	53.4/285.8	53.4/285.8	23.4/194.8
37	1.07	0.95	24.5	32 °C	5/0	4.959	98	7.8	11.9/352.3	35.2/346.2	77.0/168.2
38	0.57	19.4	0.8	23 mT	4/0	3.955	67	11.3	-48.3/127.2	-48.3/127.2	-41.2/ 8.0
39	2.08	60.8		5.0 mT	5/1	4.929	60	9.9	-32.7/168.5	-32.7/168.5	-79.1/340.6
40	1.28	52.7	0.73	3.1 mT	9/3	5.955	111	6.4	-27.9/179.6	-27.9/179.6	-85.7/260.8
41	1.10#	0.02	1.4	19 mT	11/1	10.825	57	6.1	24.3/311.3	43.2/330.4	62.0/184.2
42	1.80	6.80	14.8	71 mT	5/1	4.977	171	5.9	1.8/325.0	1.8/325.0	50.9/142.4
43	2.50	26.5	7.7	337 °C	3/1	1.998	512	11.1	11.2/322.3	-8.2/325.2	50.9/151.5
44*	1.40	4.40	6.3	226 °C	4/0	3.927	41	14.5	68.2/271.0	50.6/ 58.7	15.4/216.3
45	79.9	442.	5.4	353 °C	3/1	1.985	65	31.4	18.4/354.0	38.5/345.7	78.7/108.4
46	226.	577.	11.7	304 °C	4/0	3.763	13	26.8	-18.7/108.6	-9.2/283.9	-20.6/350.0
47*	590.	168.	112.	176 °C	3/0	2.661	6	56.2	74.9/214.3	50.0/ 30.0	-4.7/241.1
48*	842.	54.2	730.	166 °C	3/0	2.978	90	13.1	66.4/31.40	57.0/357.4	51.0/289.7
49	292.	44.8	1381.	215 °C	3/0	2.991	217	8.4	22.0/330.7	46.7/328.1	60.7/155.7
50	23.4	69.8	11.0	410 °C	3/0	2.998	928	4.0	45.1/330.1	45.1/330.1	61.5/187.5
51	2.28	6.40	10.7	162 °C	3/0	2.979	96	12.2	7.1/307.5	7.1/307.5	36.5/156.6
52	5.56	8.00	23.	475 °C	3/0	2.386	3	83.9	20.6/336.3	47.6/335.6	65.6/149.7
53	3.28	7.00	13.2	262 °C	4/0	3.770	13	26.5	56.2/334.2	56.2/334.2	61.4/210.0
54	1.37	9.70	6.2	157 °C	4/0	4.751	16	19.7	32.7/349.3	2.8/330.4	79.8/160.8
55	17.8 #	0.23	2.6	310 °C	4/0	3.954	65	11.5	37.7/334.7	18.7/338.2	66.9/145.4

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TABLE 2. (continued)

No.	NRM A/m # $\cdot 10^{-3}$	Susc. (SI) $\cdot 10^{-3}$	Q	MDF MDT		n/r	R	k	α_{95} deg	Inc/Dec tilt uncorr. deg	Inc/Dec tilt corr. deg	Lat/Long deg
56	0.57	1.20	12.5	316	°C	3/0	2.982	113	11.7	4.1/314.8	18.2/312.0	42.6/161.7
57	168.	3.08	5.7	165	°C	4/0	3.951	61	11.9	20.5/300.0	33.5/291.4	25.5/178.4
58	0.14	0.48	10.9	64	°C	3/0	2.969	65	15.4	69.1/334.7	54.9/306.8	40.6/197.4
59	8.02	79.	7.07	312	°C	3/0	2.976	84	13.6	61.7/321.7	61.7/321.7	50.0/211.7
60	79.8	148.	16.1	316	°C	3/0	2.996	533	5.3	33.3/315.8	33.3/315.8	48.2/172.8
61	0.84	0.30	83.3	351	°C	4/0	3.738	11	28.4	32.9/342.7	52.2/320.5	52.2/195.9
62	3.02	3.62	59.5	289	°C	3/0	2.882	17	31.0	33.0/318.5	25.0/304.5	36.6/169.1
63	1.10	2.19	15.2	369	°C	3/0	2.299	3	93.4	58.0/160.5	58.0/160.5	-29.4/274.1
64*	14.8 #	0.18	2.3	55	°C	3/0	2.964	55	16.7	55.8/297.6	64.2/ 83.0	18.4/303.3
65*	3.74	1.03	113.	389	°C	3/1	1.998	545	10.7	63.0/287.8	3.7/280.2	10.2/165.0
66	6.06#	0.28	0.62	73	°C	3/0	2.815	11	39.5	4.2/315.8	0.5/315.0	43.5/150.6
67	38.9 #	0.16	7.34	173	°C	3/0	2.986	139	10.5	38.8/324.3	57.2/312.7	56.6/177.7
68	3.05#	1.57	29.4	257	°C	3/0	2.996	500	5.5	12.9/310.4	20.0/304.5	40.2/158.9
69	17.1 #	0.44	1.3	188	°C	3/0	2.937	31	22.2	51.7/333.1	78.4/322.7	62.5/200.7
70	57.2 #	0.21	8.3	290	°C	3/0	2.941	34	21.5	28.9/305.2	20.4/297.0	37.7/171.7
71	43.7 #	0.29	4.28	167	°C	3/0	2.972	70	14.8	45.9/330.5	65.3/318.0	61.8/189.0
72	0.13#	0.54	6.0	224	°C	4/0	3.791	14	25.1	38.7/332.9	5.9/315.7	64.5/177.4
73*	0.38#	0.32	35.4	298	°C	3/0	2.987	150	10.1	27.4/279.1	10.1/282.5	13.1/177.7
74	0.32#	2.17	31.0	322	°C	3/0	2.999	1542	3.1	51.7/327.6	63.7/301.3	58.1/197.6
75	0.10#	1.26	2.3			4/0	3.864	22	19.9	36.5/311.2	48.7/306.9	44.2/176.7
76	0.40#	1.51	0.29			4/0	3.979	143	7.7	-49.2/158.1	-61.5/152.0	-67.3/ 19.9
77	0.15	1.35	3.0	7.3	mT	7/0	6.958	115	5.7	35.4/349.8	35.4/349.8	80.3/175.7
78	0.42	0.39	31.4	543	mT	7/0	6.992	789	2.2	14.3/143.4	14.3/143.4	-45.8/315.0
79	1.25	7.35	8.9	472	mT	12/0	11.914	127	3.9	25.4/ 0.2	25.4/ 0.2	84.4/ 76.1
80	4.35	5.00	24.2	15	mT	9/2	8.489	16	13.4	-17.6/174.1	-17.6/174.1	-78.5/288.0
81*	1.35#	0.07	0.58	29	mT	9/1	7.395	5	25.7	-39.3/274.9	-39.3/274.9	-2.7/145.7
82	3.62#	0.18	0.58	42	mT	6/0	5.975	199	4.7	-17.2/146.7	-17.2/146.7	-56.4/335.7
83	0.35	22.9	0.45	23	mT	7/0	6.978	271	3.7	27.9/313.0	27.9/313.0	44.9/170.5
84*	0.91	14.2	1.76	15	mT	7/0	6.960	142	5.1	32.5/286.6	32.5/286.6	20.7/180.2
85*	70.0 #	6.59	0.30	7.6	mT	3/1	2.873	16	32.2	37.9/ 16.0	37.9/ 16.0	74.7/333.9
86	0.13	30.1	0.12	19	mT	3/2	2.996	504	5.5	-25.6/128.5	-25.6/128.5	-40.4/350.3
87	87.9			24	mT	7/0	6.989	560	2.6	-33.8/212.0	-33.8/212.0	-59.8/155.2
88	0.12	0.38	14.3	45	mT	13/2	10.709	34	7.9	-21.3/300.0	-21.3/300.0	24.0/146.4
89	0.22	0.10	69.3	4.6	mT	6/0	5.822	28	12.9	4.9/294.5	4.9/294.5	24.0/162.8
90	2.13	22.2	2.5	>20	mT	5/1	4.990	385	3.9	36.2/342.4	36.2/342.4	73.2/179.1
91	51.0 #	4.48	0.4	>20	mT	6/1	5.935	77	7.7	-31.8/147.3	-31.8/147.3	-58.9/352.5
92	0.13	0.13	34.7	48	mT	7/0	6.870	46	9.0	-29.6/141.5	-29.6/141.5	-53.2/351.4
93	0.17	27.2	0.2	4.6	mT	7/1	6.593	15	16.3	53.5/ 33.7	53.5/ 33.7	56.0/314.0
94*	0.27	26.6	0.3	6.4	mT	7/0	6.965	170	4.6	-47.7/105.3	-47.7/105.3	-21.6/154.0
95*	59.2 #	3.68	0.47	458	°C	7/0	6.833	36	10.2	-48.2/125.5	-48.2/125.5	-39.2/ 13.9

Note: Code number of site, asterix indicates that this site w as omitted for paleopole calculation; Q, Königberger ratio; MDF, median destructive field; MDT, median destructive temperature; n/r, number of samples used/rejected for site mean calculation; R, k, α_{95}, statistical parameters (Fisher, 1953); Inc/Dec, inclination/declination of site m ean direction; Lat/Long, latitude and longitude of site mean VGP used for paleopole calculation.

Volcanic rocks behaved rather simply, as secondary magnetizations of possibly viscous origin could easlily be removed in AF amplitudes of about 10 mT. Thereafter, the trajectory of the curves is linear (Fig. 4a), even though not always directed to the origin of the diagram. This may indicate the presence of a high coercive component, possibly residing in (titano-) hematite. Such components are interpreted to be of secondary origin.

Limestone specimens responded simply to AF demagnetization (Fig. 4b). Secondary components,

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if present, could be removed in fields less than 20 mT, leaving only a stable magnetization direction which we interpret as primary.

For most specimens of intrusive rocks characteristic magnetization directions were obtained. After destruction of components which represent a large fraction of the NRM and correspond to low coercive VRM, generally stable components could be isolated (Figs. 4c to e). Some specimens showed erratic behaviour at high AF amplitudes or temperatures (e.g. Fig. 4f), hindering the determination of a characteristic direction.

Remanence Directions and Paleopoles

Site mean directions were calculated (Fisher, 1953) using sample directions obtained by individual evaluation of demagnetization curves. Directions lying far from the mean were eliminated according the criteria proposed by McFadden

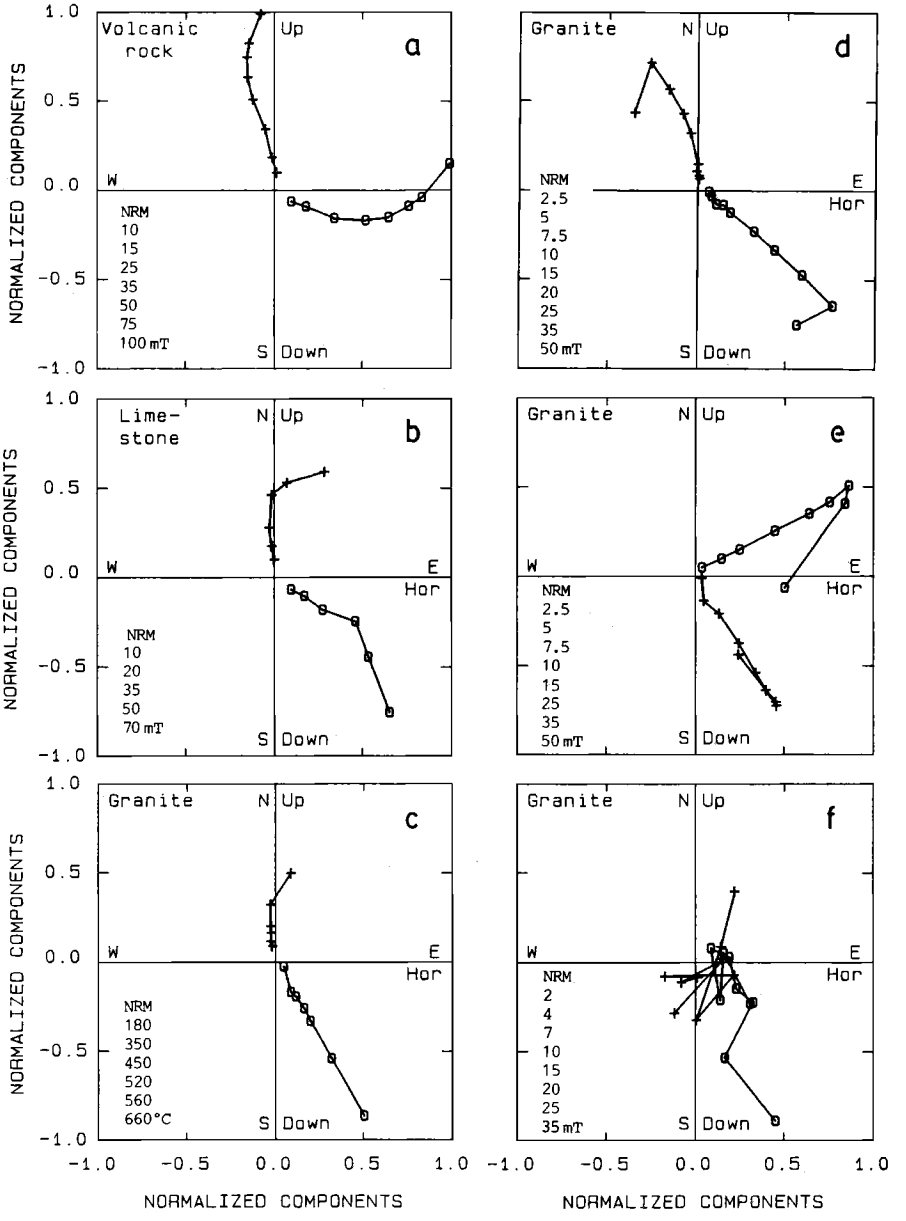


Fig. 4. Normalized orthogonal vector demagnetization diagrams (As, 1960; Zijderveld, 1967). Crosses: horizontal component; circles: vertical component.

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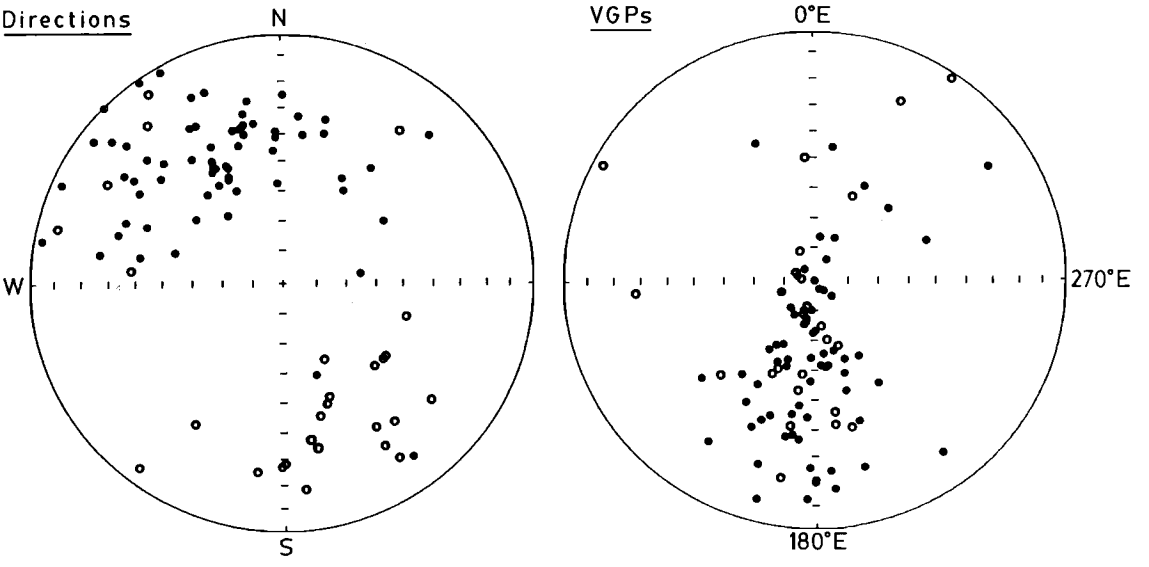


Fig. 5. Site mean directions and VGPs for all sites (equal area projection). Dots on lower hemisphere; circles on upper hemisphere.

(1980). In 3 cases no site mean direction was obtained as the individual sample directions exhibited very high dispersion and/or the sample directions became very unstable during subsequent demagnetization steps. Directions of remanent magnetization show mainly downward inclinations and northwesterly declinations (Fig. 5). The magnetic polarity is normal for about 75% of the sites. Virtual geomagnetic poles (VGP) generally coincide with the corresponding segment of the apparent polar wander path (APWP) for North America or are displaced to the left (Fig. 5). For further interpretations we calculated paleopoles for different regions and/or ages. Only VGPs from sites with k greater than 20 or α_{95} less than 20 were used, and sites with intermediate pole positions (latitude less than 20°) were rejected. Directions and VGPs for the different data sets used to calculate paleopoles are plotted in Fig. 6. The paleopoles together with statistical parameters are given in Table 3.

For the Puerto Vallarta locality (sites 1 to 22) a well defined paleopole with A_{95}=6.2° was obtained (Table 3, entry A) . No fold test is available for this data set. Theoretically, the intrusive body may have experienced tilting during the emplacement, which would produce an inclination and to a minor degree a declination anomaly of the paleomagnetic data. As we did not observe large inclination anomalies for this intrusive complex nor for the other intrusive bodies along the study area, we suppose that tilting here is not an important factor. The polarity of remanence for the Puerto Vallarta locality is predominantly normal: only 5 of 22 sites exhibited reversed polarity. In regard to the age of these rocks between 105 Ma and 80 Ma, this corresponds well to the polarity time scale, as most of this age range belongs to the Cretaceous Normal superchron (e.g. Harland et al., 1982), which ends 83 Ma ago.

Entry B (sites 25 to 40) represents Early Tertiary volcanic and intrusive rocks. The age range may further be restricted, as all geochronological dates point to ages between about 35 Ma and 65 Ma (Pantoja-Alor, 1983, 1988, and personal communication; own data). In this sampling region some of the volcanic units (sites 25 to 31, and 37) requiered tilt corrections, indicated by the dips of underlying sediments. The tilting of these rocks probably is related to the numerous intrusions along the margin. The structural correction reduces dramatically the between-site dispersion: for 8 sites the precision parameter K increases from 8.5 to 42.8, which is significant at the 99.5% probability level. In contrast, the position of the mean pole remains almost unaffected (193°E/74°N before and 162°E/72°N after correction). Using the tectonically corrected and uncorrected VGPs together with VGPs from the same area where no structural correction was possible, the fold test is still significant at the 95% probability level.

Sediments sampled in the Encino mining district (Table 2, entries 66 to 76) probably were remagnetized as indicated by a positive fold test. For 10 sites the precision parameter decreases from 31.0 before to 13.5 after structural corrections, which is significant at the 95% probability level. The mean pole position again is not much affected (176°E/55°N before and 195°E/45°N after structural correction). The remanence of the sediments probably was reset by the emplacement of a large intrusive body during the Maastrichtian. This is probably not the case for Albian-Aptian volcanic and intrusive rocks from the same sampling locality (Table 2, entries 55 to 65). For these rocks

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tilt corrections were applied as measured for the underlying sediments. The tilting of both the igneous and sedimentary rocks is probably due to the nearby intrusion and to some extent to the regional folding. Unfortunately, the effects of these two tectonic processes could not be separated. The precision parameter for the VGP dispersion increases from K=13.1 to K=19.9 after structural correction, which is not significant at the 90% probability level. Probably the remanence of the volcanic rocks was more stable than in the sediments and resisted a remagnetization. The corresponding mean pole positions are very close to each other (187°E/50°N and 180°E/47°N, respectively). For further interpretations we will combine the VGPs from the unremagnetized igneous rocks to an Albian - Aptian paleopole, and the VGPs from the Maastrichtian intrusive and Senonian rocks to a Senonian paleopole (Table 3, entries C and D). The Albian-Aptian volcanic rocks from the Encino area all show normal polarity of magnetization, in full agreement with the polarity time scale.

Tertiary igneous rocks (ca. 35 Ma to 65 Ma old, see above) from the Arteaga-Zihuatanejo locality (sites 80 to 93) are concluded in paleopole E. The rocks present mixed polarity with 7 normal, 6 reversed, and 1 intermediate VGPs.

All paleopoles discussed so far correspond to the Guerrero terrane (see Fig. 2). The Xaltianguis and Acapulco intrusives (Table 3, entries F and G) belong to the Xolapa terrane. These granites were emplaced about 30 Ma and 43 Ma ago. Paleomagnetic results are discussed in Urrutia-Fucugauchi (1983) and Böhnel et al. (1988). For paleopole calculation we used only sites meeting the criteria given above.

The last paleopole (Table 3, entry H) represents the Albian-Cenomanian Morelos Formation north of Acapulco (see Fig. 2b) and was reported by Urrutia-Fucugauchi (1988). This locality is situated within the Mixteco terrane.

In regard to possible tectonic movements, the paleopoles for the study area were compared with the corresponding reference poles for stable North America given by Irving and Irving (1982) (10 to 90 Ma) and Harrison and Lindh (1982) (100 to 120 Ma) (Fig. 7; Table 3). Where neccessary, geological ages were transformed to absolute ages according to Harland et al. (1982). Following Beck (1988) and Demarest (1983), apparent rotations (R) and poleward displacements (P) were calculated.

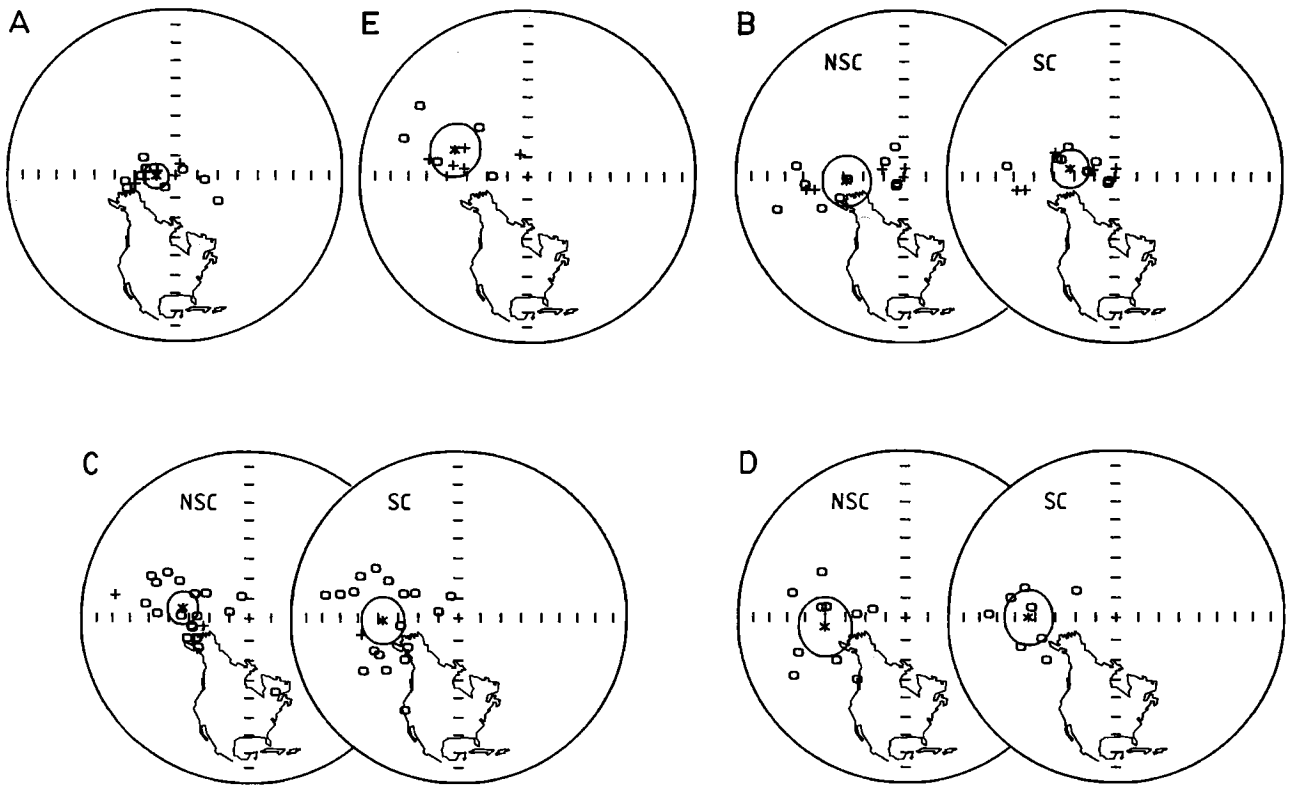


Fig. 6. Site mean VGPs (dots and crosses) and mean poles (stars) with 95% confidence limits for different sampling regions in southern Mexico. Letters A to D correspond to Table 3. NSC: not structurally corrected, SC: structurally corrected. Circles and stars on lower hemisphere, crosses on upper hemisphere (equal area projection).

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TABLE 3. Paleopoles for Sampling Areas in Southern Mexico

	Region	Sites	Age Ma	N	R	K	A95 deg	Lat/Long deg	Paleolat. deg	R deg	P
A	Puerto Vallarta	1-22	80-100	14	13.69	41.4	6.2	80.7/182.7	23 ±6	13 ±8	4 ±10
B	Autlan	25-40	40-70	14	13.32	19.2	9.3	68.2/169.5	20 ±9	-12 ±15	4 ±11
C	Encino	43-54, 66-76	65-83	19	18.07	19.4	7.8	57.0/171.2	18 ±8	-15 ±12	7 ±9
D	Encino	55-63	98-119	8	7.65	19.9	12.7	46.8/180.2	21 ±12	-24 ±13	1 ±13
E	Arteaga-Zihuat.	80-93	35-65	10	9.41	15.4	12.7	52.1/159.8	9 ±13	-27 ±17	12 ±14
F	Xaltianguis	96-100	30	3	2.89	18.4	29.6	68.9/178.0	19 ±30	-18 ±25	-3 ±24
G	Acapulco	101-116	43	12	11.14	12.8	12.6	81.0/199.6	21 ±13	-2 ±11	-5 ±11
H*	Morelos Form.	117	91-113	28*	27.87	215	3/5*	62.9/197.7	28 ±3	-3 ±10	-9 ±12

Note: Site numbers refer to Tables 1 and 2. Age range for sites; N, number of sites used for paleopole; R, K, A95, statistical parameters (Fisher, 1953); Lat/Long, latitude/longitude of paleopole; paleolatitude for sampling area; R, P, apparent rotation (positive clockwise) and poleward displacement (positive northward) according to Beck and Demarest (1983); reference poles were used for the age range in column 4. *: this entry represents one site only; here, N gives the number of samples and A95 the axes of the confidence oval around the site mean VGP.

The errors of R and P were calculated considering the errors of observed and expected paleopoles, as well as the age uncertainty for the observed poles, by comparing with reference poles covering the whole eage range. All but one paleopole show no significant poleward displacement, indicating that the Guerrero, Xolapa, and Mixteco terranes have not experienced any latitudinal movement relative to stable North America since about 120 Ma. This follows also from Fig. 8 where the paleolatitudes calculated from the paleopoles are plotted vs time, together with the paleolatitudes calculated from the reference poles for the coordinates of Zihuatanejo. A plausible explanation, at least for the Guerrero and Xolapa terranes, is an origin near the continental margin followed by minor movements. This would be in agreement with tectonic models for the Guerrero terrane, in which the composition and distribution of rocks are related to an island arc and a fore arc or back arc basin, which was closed later by changes in the plate configuration (Campa and Ramirez, 1978; Urrutia-Fucugauchi and Valencio, 1986). One could expect that the closure of such a basin would produce large deformations and differential block rotations. As mentioned above, only moderate deformation is reported for the Guerrero terrane. Additionally, most of the paleopoles (see Table 3) indicate no significant apparent rotations. The Puerto Vallarta paleopole (entry A in Table 3) indicates a small clockwise rotation, which partly could have resulted from movements of the Rivera plate, separating since few Ma from the continent. Declination anomalies for the Encino (entries C and D) and Autlan (entry B) areas probably indicate counterclockwise block rotations which may be due to left-lateral shear. Such shear could be related to oblique plate convergence which occurred frequently along the Mexican continental margin (Fig. 11). Alternatively, the anomalous declinations could be explained by multiple non-coaxial deformation phases.

In view of this complex situation, we consider the probable interactions of oceanic plates with the Mexican margin according to models for the tectonic evolution of the Pacific Basin. Most plate tectonic models for the Pacific Basin indicate the interaction of the Farallon and/or Kula plates with North America over most of the past 180 Ma, and of the Pacific plate only since the last few million years. Engebretson et al. (1985) developed a model assuming that hot spots in the Atlantic were fixed relative to hotspots in the Pacific region. According to this model, the only plate interacting with North America prior to 80 Ma was Farallon. Between 180 and 125 Ma the convergence involved left lateral and thereafter right lateral components. At 80 Ma the Kula plate came into contact with the continental margin. The southern extent of this contact is unknown due to

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uncertainties in the position of the Kula - Farallon spreading axis (Fig. 9a). Possible locations of the spreading axis are somewhere between northern California (northern option) and Guatemala (southern option). The northern option would place the Farallon plate adjacent to western United States from 85 to 56 Ma, producing rapid plate convergence that may have been associated with Laramide tectonism. The southern option places the Kula plate along the southern North American continental margin, offering a mechanism for rapid northward transport of terranes. At 56 Ma the Kula plate had moved up to northern North America and the Farallon plate again was subducting beneath Mexico and Central America. Using this model, Debiche et al. (1987) calculated trajectories of terranes transported by the Farallon and Kula plates (choosing the southern option for the Kula - Farallon spreading ridge), as well as maximum coastwise translation of terranes due to oblique subduction episodes.

In Fig. 9b trajectories are shown for hypothetical terranes in fixed North American coordinates as determined by backward modelling for different arrival times at the Mexican continental border. The trajectories indicate a large scale northward displacement for a 60 Ma arrival, but almost eastward displacement for 120 Ma arrival. After dokking, coastwise transport may have occurred, according to Debiche et al. (1987) predominantly in a northerly direction. The trajectories may be translated into paleolatitude vs time curves, and Fig. 10a shows that terranes arriving 60 Ma ago at the southern Mexico continental border exhibit latitude displacements relative to North America of less than 10°, with much smaller values for arrival times of 90 Ma and 120 Ma.

Jurdy (1984) reconstructed the subduction of the Farallon plate beneath North America since the late Mesozoic using relative plate motion data (Fig. 10b). For times before 60 Ma and after 40 Ma trajectories resemble those given by Debiche et al. (1987), indicating northward displacement of Farallon relative to North America, but from 60 Ma to 40 Ma this movement was almost eastwards. During this short period large left displacements may have occurred, as the convergence involved southward tangential components exceeding 100 km/Ma (Fig. 11). This would offer a mechanism for a coastwise transport of the Chortis Block to Central America, as proposed by Gose and Swartz (1977). Fig. 11 further shows convergence data from Engebretson et al. (1985), indicating that also before that period left lateral tangential displacement was probably more frequent along the southern Mexico continental border than right lateral displacement. This then would offer a mechanism to move terranes situated since at least the Early Cretaceous in southern Mexico (e.g. Mixteco) southwards, as proposed by Moran-Zenteno

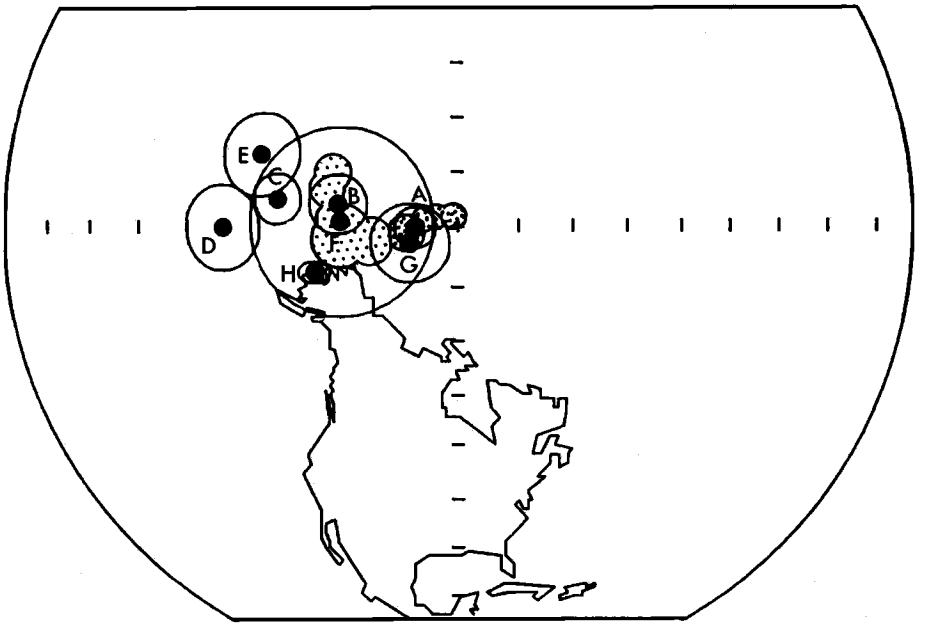


Fig. 7. Paleopoles for different sampling regions with 63% error limits (equal area projection). The Cretaceous and Tertiar-to-Recent segment of the apparent polar wander path for North America with 63% error limits according to Irving and Irving (1982) (10-90 Ma) and Harrison and Lindh (1982) (100-150 Ma) is shown shaded and heavy shaded, respectively. Letters A to H refer to Table 3.

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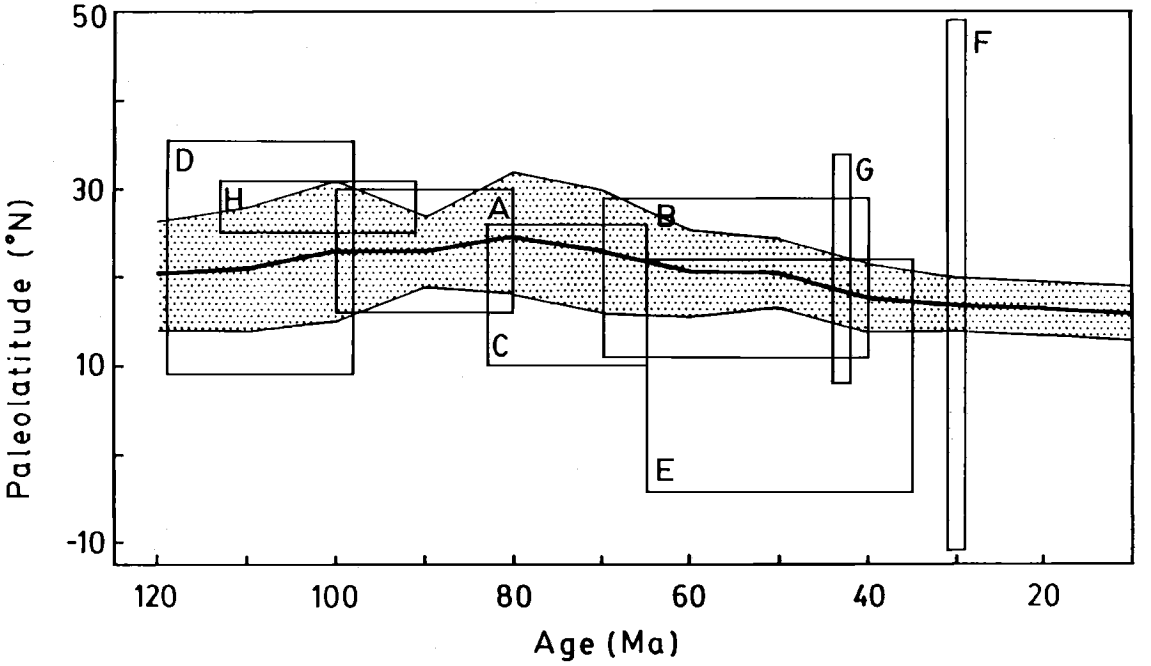


Fig. 8. Expected paleolatitude vs time curve for Zihuatanejo (see Fig. 2), assuming that this point was fixed relative to cratonic North America. The thick line represents the paleolatitude calculated from Irving and Irving (1982) (10 Ma to 90 Ma) and Harrison and Lindh (1982 (90 to 120 Ma) with 95% error limits shown by the shaded area. Each rectangle is defined by the 95% error bars of paleolatitude and the age range for the rocks concluded in the paleopole (see Table 3).

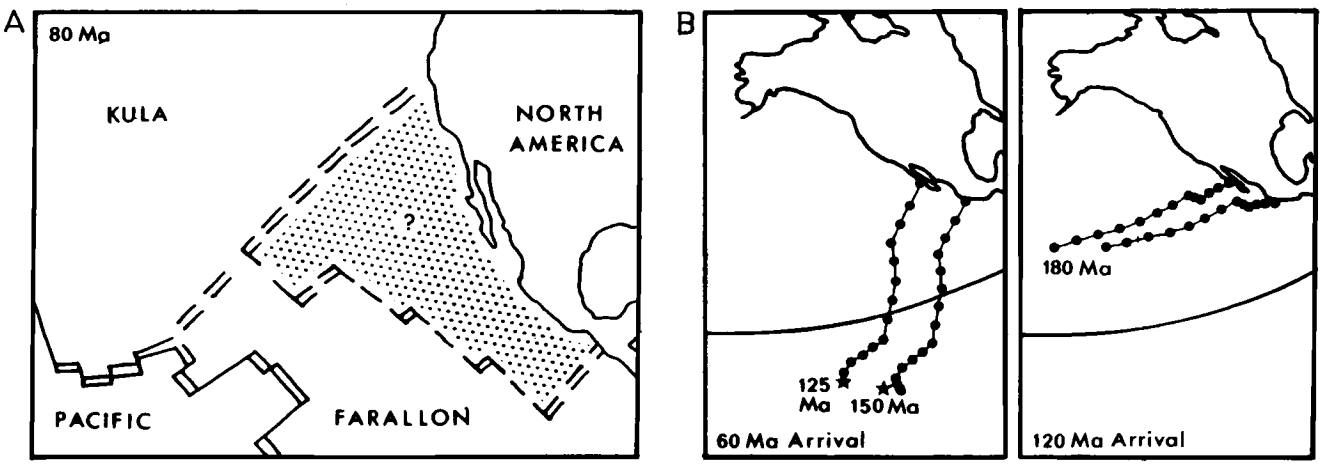


Fig. 9. (a) Configuration of plates in the Pacific Basin at 80 Ma before present according to Engebretson et al. (1985). The shaded area could belong either to Kula or Farallon plate. (b) Trajectories for hypothetical terranes in 10 Ma intervals for arrival times of 60 Ma and 120 Ma at the Mexican border (Debiche et al., 1987). Stars mark positions on spreading axes. Paleoequators are shown by convex curves.

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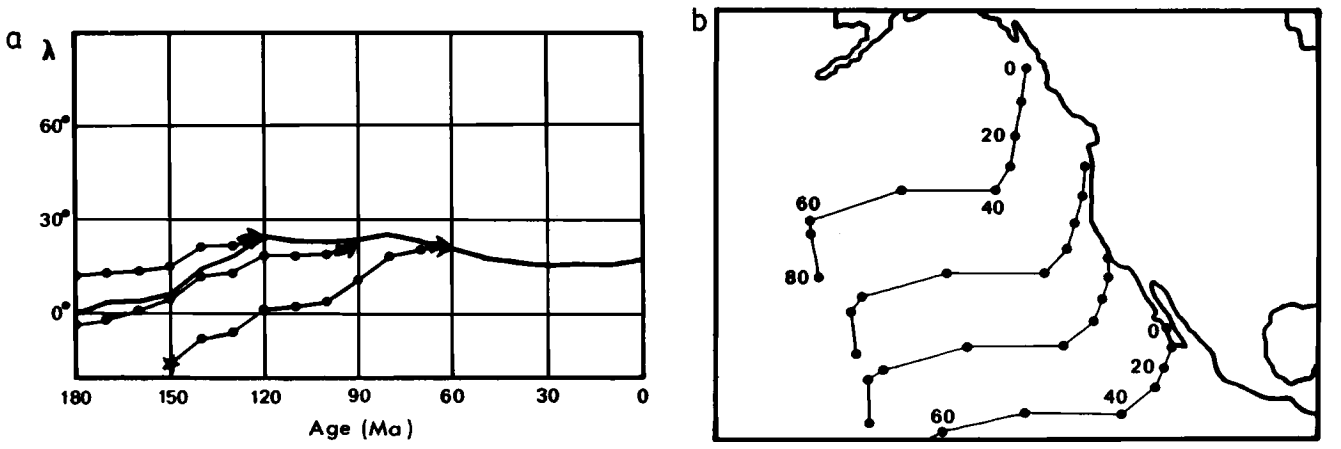


Fig. 10. (a) Paleolatitude vs time curves for a point along the Mexican border (thick line) and for three hypothetical terranes arriving at 120, 90, and 60 Ma at the continental margin. The star indicates a position on a spreading axis. (b) Trajectories of 4 points on the Farallon plate as derived from relative plate motion data (Jurdy, 1984), relative to North America.

(1987). Such southward transport of terranes probably also occurred along the western continental margin of Canada and the USA in times before about 125 Ma, but after that time only displacements of terranes to the north are reported (Beck, 1988).

We may conclude that according to plate tectonic models for the Pacific Basin, southern Mexican terranes should not show large changes in paleolatitude relative to North America. According to uncertainties in the position of the Kula - Farallon spreading axis and differing convergence directions in different models, the possible amount of coastwise transport of terranes is not well defined. Southward displacement probably occurred frequently, but periods of northward displacement can not be excluded and are even strongly favoured according to data from western North America (e.g. Beck, 1988).

Beck (1983) proposed that oblique convergence was the most important mechanism for the northward transport and clockwise rotation of terranes now situated along the northern North America Pacific margin, and that possibly these terranes originally were situated near southern Mexico. Our data suggest that that the Guerrero, Xolapa, and Mixteco terranes did not participate in such movements. Indeed, our data suggest that left lateral and southward displacements have affected the Mexican border since about 60 Ma. Such movements agree well with plate tectonic models for Central America and Mexico (e.g. Anderson and Schmidt, 1984) and paleomagnetic data for the Chortis Block (Gose and Swartz, 1977). This points to a major discontinuity in the tectonic processes along western North America, somewhere north of the Guerrero terrane. One possible explanation could be that the spreading axis between the Kula and Farallon plates was situated to the north of the terranes studied here, allowing northward and southward terrane transport, consecutively or even at the same time. In this case the spreading axis should be placed more or less between the southern and northern option proposed by Engebretson et al. (1985) (see Fig. 9a). Possibly rather high spreading rates between Kula and Farallon are required to displace terranes to their present positions in

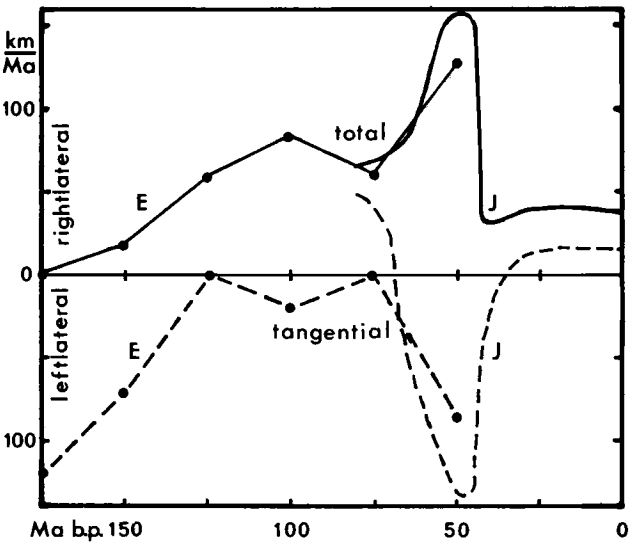


Fig. 11. Total and tangential convergence rates from 175 Ma to present along the southern Mexico continental margin, recalculated from Engebretson et al. (1985) (E) and Jurdy (1984) (J).

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Central America and along the Pacific margin of northern North America. Other explanations may involve changes in the so far proposed plate configurations in the Pacific Basin, possibly including additional plate boundaries and/or different evolution in time.

On the other hand, an origin of the Guerrero (and possibly the Xolapa) terrane in a position relative to Mexico similar to the present one would restrict plate recontructions of Pangea, in the fit between South America and North America (see e.g. review in Urrutia-Fucugauchi, 1984). If the oldest rocks within the Guerrero terrane are Lower Jurassic as reported by Grajales-Nishimura and Lopez-Infanzon (1983), South America would have to be placed near the Mexico - Guatemala border, to avoid overlapping.

Conclusions

The southern Mexico continental margin is characterized by Cretaceous-Early Tertiary calc-alkaline batholiths and volcanics emplaced in a sedimentary and metamorphic terrane (Fig. 1). The margin has experienced a complex evolution involving plate subduction, accretion of terranes to the margin and lateral transform faulting resulting in margin truncation (e.g. Malfait and Dinkelman, 1969; Karig et al., 1978; Beck et al., 1981; Urrutia-Fucugauchi, 1983). The paleomagnetic results indicate that this part of the margin has not experienced large-scale rotations or significant latitudinal translations since the mid-Cretaceous. This supports previous results reported for the Puerto Vallarta batholith complex (Böhnel, 1985; Böhnel and Negendank, 1988) and for the Acapulco and Xaltianguis granitoids (Urrutia-Fucugauchi, 1983; Böhnel et al., 1988a), which lie to the northwest and southeast, respectively, of the area described in this paper. Additionally, paleomagnetic results for Albian-Cenomanian limestones from Chilpancingo (Trevino-Rodriguez, 1986; Urrutia-Fucugauchi, 1988), Petlalcingo (Böhnel, 1985) and Teposcolula (Jurado-Chichay, 1988), all from the Mixteco terrane, generally indicate tectonic stability of these areas of southern Mexico, except for localized rotations (see e.g. Böhnel, 1985; Böhnel et al., 1988). Such rotations possibly are related to left-lateral shear along the continental margin.

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